

MINAHIL MOBIN

MS Cybersecurity Student • AI/ML • Generative AI • Cloud Security
Lahore, Pakistan • +92 325 4112172 • minahil.aj.work@gmail.com • GitHub • LinkedIn • Portfolio

Education

MS Cybersecurity (In Progress)	2025 - 2027
National University of Computer and Emerging Sciences (FAST-NUCES), Lahore	CGPA: 3.17
BS Data Science	2021 - 2025
National University of Computer and Emerging Sciences (FAST-NUCES), Lahore	

Experience

AI/ML Intern, Abacus (In Progress)	July 2026 - August 2026
- Developed Python applications using object-oriented programming, file handling, exception handling, and modular design. - Built automation tools for document processing, text analysis, and JSON report generation. - Performed data cleaning, exploratory data analysis (EDA), and visualization using Pandas and Matplotlib. - Implemented and evaluated machine learning models including Linear Regression, Logistic Regression, Decision Trees, and K-Means using Scikit-learn. - Built an AI Knowledge Assistant using LLM API integration, prompt engineering, embeddings, ChromaDB, semantic search, RAG-based document question answering, and a FastAPI backend with a basic AI agent workflow.	
MIS Intern, Shaukat Khanum Memorial Cancer Hospital	June 2024 - July 2024
- Researched healthcare information systems and emerging AI applications in clinical environments. - Explored the use of Large Language Models (LLMs) for clinical documentation assistance and medical information retrieval. - Evaluated healthcare datasets and reviewed current research to assess their suitability for machine learning applications.	

Selected Academic Projects

Explainable AI-Based Intrusion Detection System for Cloud Security	May 2026
- Developed an Explainable AI-based Intrusion Detection System (IDS) using Decision Tree, Random Forest, and SVM classifiers on the CICIDS2017 dataset. - Selected Random Forest as the best-performing model and deployed it as a Flask REST API on AWS EC2 for real-time intrusion detection. - Integrated SHAP for model interpretability and evaluated robustness through cross-dataset and noise-based testing, highlighting generalization challenges in cloud environments.	
Feature Selection-Based Lightweight Intrusion Detection System	May 2026
- Designed a lightweight Random Forest-based IDS using Mutual Information feature selection on the CICIDS2017 dataset. - Evaluated seven feature subsets and achieved 99.82% classification accuracy using only 15 of 78 network traffic features (80.8% feature reduction). - Analyzed the trade-off between feature reduction, model complexity, and detection performance for efficient intrusion detection.	

Dyslexia Support Hub (AI Learning Platform)	August 2024 - May 2025
- Developed a full-stack AI-assisted learning platform using React and Django to support personalized education for children with dyslexia. - Built adaptive learning modules and integrated machine learning techniques to generate personalized learning recommendations based on student progress.	

Inventory Management System	February 2023 - May 2023
- Developed a full-stack inventory management system using React and JavaScript for efficient product tracking and management. - Implemented CRUD operations, input validation, and structured workflows to improve data management and system reliability. - Managed project planning, documentation, and development workflow throughout the software lifecycle.	

Technical Skills

Programming: Python, JavaScript, SQL, C++
AI/ML: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, SHAP, Model Evaluation
Generative AI: LLMs, RAG, Prompt Engineering, Embeddings
Cybersecurity: IDS, Network Security, Cloud Security, Security Analytics, Wireshark
Tools: React, Django, REST APIs, AWS EC2, MySQL, PostgreSQL, Git

Certifications

AWS Academy Graduate - Cloud Security Builder	May 2026
AWS Academy Graduate - Cloud Security Foundations	April 2026
AWS Academy Graduate - Cloud Foundations	April 2026